

Summary for Generating ELM Surface Dataset

Overview of current workflow in order

Related utilities

Other spatial/temporal input files (namelist items)

Plan for updating land surface data

PFT/CFT update - short to medium term

Topounit update - medium term

Demeter-mksurldata - medium term

Comprehensive land data system - long term

Project priorities

Current workflow

1. Grid Files

2. Mapping files

3. Domain files

4. PFT and CFT data files

No crop model - PFTs only

Crop model - PFTs and CFTs

5. ELM land surface files

mksurldata

No crop model - PFTs only

Crop model - PFTs and CFTs

With topounits

Current mksurldata data files (mksurldata namelist), v3.0.2

Desired new mksurldata data files

Overview of current workflow in order

1. Grid files

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5. ELM land surface files

Related utilities

- Regridding initial ELM files: E3SM/components/elm/tools/interpinic

Other spatial/temporal input files (namelist items)

- scenario dependent
 - nitrogen deposition

- population density
- phosphorous deposition - spatial only

Plan for updating land surface data

- Workflow is based on tools that run separately from E3SM
- PFT/CFT workflow will be duplicated for online E3SM-GCAM code
- Workflow can be automated via shell (or other) script
 - except for adding new resolutions and mapping files to E3SM/cime_config/config_grids.xml
 - updating this text could probably be automated, but this file is in the E3SM git codebase
- New meshes are in the works (See [Unified MPAS meshes](#))
 - this will change the grid files and the mapping files
 - will mksurdata be able to do the new grid conversions?
- Need to integrate fine-res land cover data with fine-res land use data
 - need historical time series
- Need a PFT/CFT update - short to medium term
 - new reference and land use/cover data
 - for crop model - CFTs
 - for E3SM-GCAM - complete GCAM data
 - for FATES - transitions
 - fine-resolution - will need parallelization
 - 1km land cover 2001-2020 (Li et al, ESSD, 2024)
 - MOD44B - vegetation continuous fields 2000 - present (250m)
 - percent tree-cover, non-tree-cover, bare
 - MODIS land cover/PFT at 500m (MCD12Q1) and at 1km (MCD12Q2)
 - LANDSAT global forest change 2000-2024, global land cover 2000-2020, 30m
 - ESA-CCI data
 - land cover, 300m, 1992-2015
 - PFT, 300m, 1992-2020
 - water bodies, 150m, static circa 2000-2012
 - ESA worldcover, 10m, 2020-2021
 - Copernicus land monitoring service, land cover 2015-2020, 100m - 10m
 - GLC_FCS30D - 30m 1985-2022

- GFSAD has static, limited global crop detail at 30m and 1km
- [MapSPAM](#) has 46 crops with four management intensities at 5 arcmin (FAO-based)
 - physical area, harvested area, production yield (at least for 2020)
 - 2000, 2005, 2010, 2017, 2020
- wood harvest: FAO data not gridded, LUH data at quarter degree
- !!!!!!!Need to stop replacing shrubbery with forest in the tropics!!!!!!
- Store only potential land cells in the files
- Need to include topounits - medium term
 - single topographic data input to mksurfddata
 - what is the relationship between the top radiation scheme and topounits?
- Demeter-mksurfddata workflow - medium term
 - for E3SM-GCAM
 - Can incorporate all of the above
- Comprehensive land data system - long term

PFT/CFT update - short to medium term

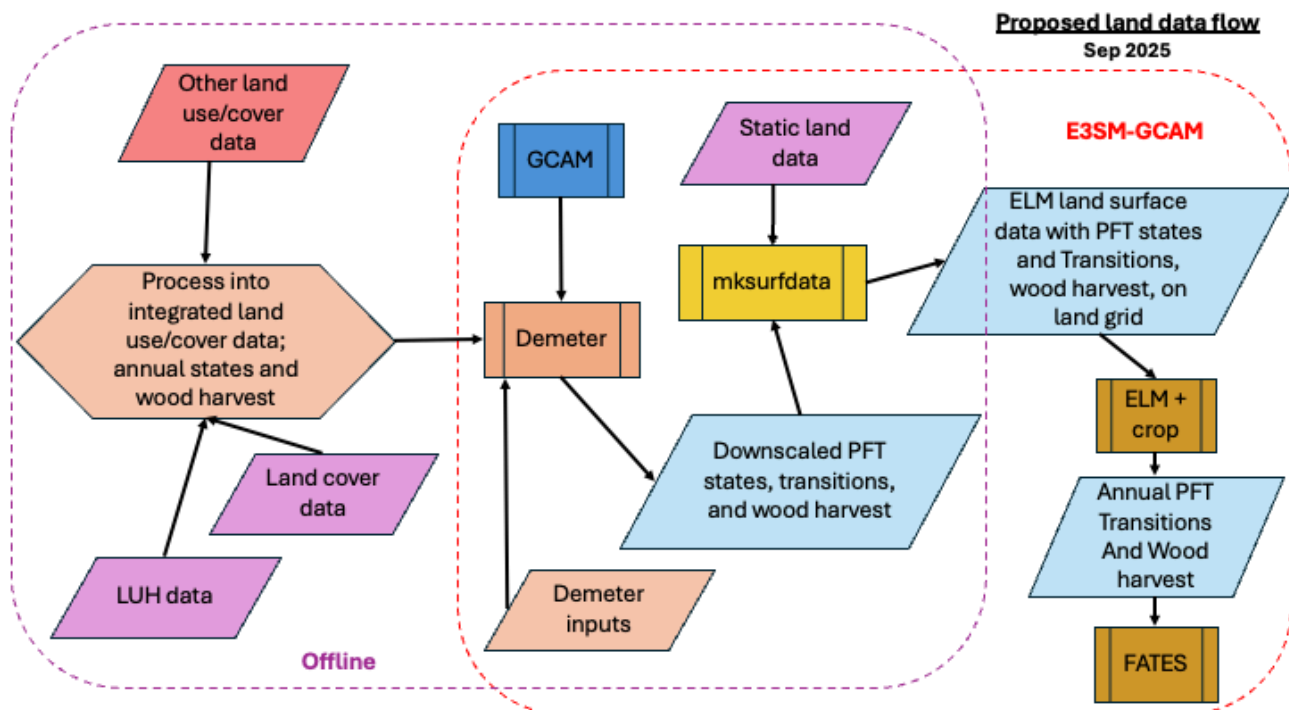
- LUT expansion to include irrigated crop pft (no crop model) and/or CFTs for the crop model
 - crop data source?
- Reference (year 2000) PFT and land use files for LUT should be updated
 - currently 17 PFTs (irrigated crop unused) and LUH1 data
 - need 50 PFTs and updated land use data
 - currently half-degree; could be quarter degree (which is LUH2 res)
 - a different reference year could be used
- Rebuild LUT to use LUH2 data with transitions? medium term
 - Alternative data sources?
 - Should still support LUH2 and future LUH data, regardless
- mksurfddata expansion to include dynamic urban, wetland, glacier, lake?
 - how are peat data used in the model?
 - FATES needs transitions
- stop changing PFTs in mksurfddata!

Topounit update - medium term

- Topounit to be a standard sub-grid dimension for variables in land surface files
 - one topounit is equivalent to grid cell level

- one fine-resolution topography input (1km?) to mksurfddata
 - determines topounits
 - determines all other related topo-related variables
- **requires fine-res (1km) inputs to mksurfddata**
- land variables with topounit specificity
 - all variables?
 - lai, sai, canopy height, soil properties (clay, sand, organic matter), land units, pft/cft, others?

Demeter-mksurfddata - medium term



- **Use Demeter to determine land use/cover inputs for mksurfddata**
 - annual states and transitions, downscaling to grid
 - including irrigation, fertilizer, wood harvest, detailed land use and cover
 - including urban, wetland, glacier, and lake (peat?)
 - specify mappings between input, base, and output land categories
 - need to determine best approach for wood harvest data
 - baselayer determines output grid and starting point
 - can run forward or back in time
 - currently testing 150m output res for texas
 - mksurfddata will do final ?grid conversions? and surface file generation

- and incorporate the static physical characteristics
- need to parallelize?
- can pre-process different source data for Demeter
 - source data requires pre-processing and surface file generation (new LUT)
 - GCAM - regionXbasin, future
 - Moirai - countryXbasin, historical; GCAM initial data
 - needs land cover detail; currently single cover at 5 arcmin
 - crop detail present, but not currently output on the grid
 - over 100 FAO crops, calibrated to desired FAO stat year, but the physical cropland area is currently static at year 2000
 - currently no wood harvest data
 - LUH2 - quarter degree land use; need to merge with land cover
 - fine-res source data
 - 1km land cover 2001-2020 (Li et al, ESSD, 2024)
 - GFSAD has static, limited global crop detail at 30m and 1km
 - [MapSPAM](#) has 46 crops with four management intensities at 5 arcmin
 - physical area, harvested area, production yield (at least for 2020)
 - 2000, 2005, 2010, 2020
 - technically will do gridded input data, but needs to be tested
 - Demeter has processor parallelization built-in, and can be further split spatially across multiple nodes
- topounits
 - mksurldata will ingest fine-res topography data to determine topounits
 - demeter output will be fine-res
 - mksurldata will calculate topounit-specific data for relevant variables
 - non-land use/cover variables input to mksurldata will also need to be fine-res

Comprehensive land data system - long term

- Integrate all relevant land data so they are self-consistent
- perform historical reconstruction and future projection
- fine-resolution output ($\leq 1\text{km}$); 5 arcmin is readily doable now
 - this would provide self-consistent inputs for a revised mksurldata
- detailed, comprehensive land type categories
- annual states and transitions

- readily updatable/advancable as new data become available
- Users should only have to aggregate types and resolution for global applications; no disaggregating/downscaling of data
- three step approach:
 - contemporary data integration (~1970 to present) to get complete data set
 - integrated land use/management/cover and associated characteristics
 - ML approach for determining local/regional land transition probabilities from contemporary data
 - ML approach for historical reconstruction or future projection, based on contemporary data, land use reconstruction (e.g. HYDE), and transition probabilities

Project priorities

- What are the immediate needs?
- What are the medium term needs?
- What are the long term needs?

Current workflow

1. Grid Files

- Obtain or create grid file (scrip or g file) for desired resolution, and also for each source data resolution
 - .../inputdata/share/meshes/homme
 - .../inputdata/lnd/clm2/mappingdata/grids
 - inputdata/ocn/maps
 - To create a grid file see Prototypical Regridding III or IV at: [Regridding E3SM Data with ncremap](#)
 - The new unified MPAS meshes will be stored as grid files?
 - Old script for creating grid files: E3SM/components/elm/tools/mkmapgrids

2. Mapping files

- Generate mapping files to desired resolution, based on grid files
 - .../inputdata/cpl/gridmaps/<res>
 - Note that <res> is the source resolution of the mapping files in the directory
 - [Mapping file algorithms and naming convention](#)

- Land grid mapping files should use the area conserving method
 - I think all mapping files should use the area conserving method
 - The atmosphere does not always play nice with area conserving mappings
 - Recent recommendations and info:
 - [Recommended Mapping Procedures for E3SM Atmosphere Grids](#)
- ncremap appears to be the currently preferred tool (available in E3SM unified)
 - See Prototypical Regridding II at: [Regridding E3SM Data with ncremap](#)
 - ESMF_RegridWeightGen or Tempest can be called by ncremap
 - [Full ncremap documentation](#)
- Old script that set up the creation of a set of mapping files:
E3SM/components/elm/tools/mkmapdata
- The source res of these mapping files and how many are needed are determined by the data files needed by mk surfdata
- The desired resolutions and these mapping files need to be set in:
E3SM/cime_config/config_grids.xml

3. Domain files

- Generate domain files at desired lnd/ocn resolutions, based on grid files and a mapping file
 - E3SM/tools/generate_domain_files/generate_domain_files_E3SM.py
 - .../indata/share/domains

4. PFT and CFT data files

- These are the time-varying data
- Data input
 - LUH2 state and harvest data are converted to LUH1 format
 - Year 2000 reference LUH land use
 - Year 2000 reference PFT land cover
- ~~Potential updates~~
 - ~~Use all relevant LUH2 data, including transitions~~
 - ~~An updated reference PFT dataset can be created~~
 - ~~Use other sources of land use and land cover data~~
 - HYDE is the original land use source

- ~~Can utilize time series of land cover~~
- ~~Integrated land use/cover time series can be created~~

No crop model - PFTs only

- No crop land unit
- 17 PFTs with one generic, non-irrigated crop and the last, irrigated crop index is empty (includes bare PFT)
 - See [Creating land surface datasets for E3SM](#)
 - Pre-process land use data (LUH2 to LUH1 format)
 - Run land use translator (LUT) to create annual input files for mksurfddata
 - E3SM/components/elm/tools/land_use_translator
 - The files are stored in a distinct folder for each set of files, in `.../inputdata/lnd/clm2/rawdata`
- An additional text file needs to be created that lists the annual files for use in creating a landuse.timeseries file; it is stored in the same directory as the annual files
- To generate a new 1850 starting point an extra run of the LUT is required before the time series is processed.
- 17 PFTs with a generic non-irrigated crop and a generic irrigated crop
 - First use mksurfddata to create a 16 PFT file, then manually split the cropland based on ancillary data
 - Is there another way that this is currently done?
 - e.g., the LUT output file could be updated to include an irrigated crop value in the last PFT index

~~Crop model - PFTs and CFTs~~

- ~~with Crop land unit~~
- ~~up to 51 PFTs with specific crops (includes bare PFT)~~
 - ~~15 nat veg PFTs go onto nat veg land unit (bare + 14 PFTs)~~
 - ~~36 CFTs go onto crop land unit, including the two generic crops (non-irrigated, irrigated)~~
- ~~CFT data sourced from NCAR surfddata/landuse.timeseries files~~
- ~~surfddata file~~
 - ~~See [Creating surface data file with increased PFTs \(50\)](#)~~
 - ~~Two input files specific to these PFTs have been generated for use by mksurfddata:~~

- `mksrf_fvegtyp =`
`'/qfs/people/sinh210/wrk/user_inputdata/ln2/clm2/surfdata_map/mksrf_newcrop_c200813.nc'`
 - This is an LUT output file that the CFTs have been added to
- `mksrf_flai =`
`'/qfs/people/sinh210/wrk/user_inputdata/ln2/clm2/surfdata_map/mksrf_lai34cft_simyr2000_c200622.nc'`
- `landuse.timeseries` file
 - See [Creating landuse timeseries for 50 PFTs](#)
 - PFT/CFT LUT style files have not been created for use by `mksurfdata`
 - Instead, previously made `landuse.timeseries` files have been modified to include the CFT data

5. ELM land surface files

`mksurfdata`

- `E3SM/components/elm/tools/mksurfdata_map`
- Requires mapping files and PFT/CFT files, and the data files listed below
- Still changes temperate tree and shrub in the tropics to tropical tree
 - this dramatically inflates the tree/forest cover
- See `E3SM/components/elm/tools/mksurfdata_map/src/mksurfdat.F90` for full list of namelist items
- See `E3SM/components/elm/bld/namelist_files/namelist_defaults_tools.xml` for namelist defaults
- One PFT/CFT params file can be used for all land model configs:
`clm_parans_c230517_phs_50pfts.nc`

No crop model - PFTs only

- See [Creating land surface datasets for E3SM](#)
 - Section: Generating an ELM surface dataset with `mksurfdata`
- Build `mksurfdata`

- Run in “debug” mode to create the namelist
- Modify the namelist to set the desired configuration
 - There are some defaults in
E3SM/components/elm/bld/namelist_files/namelist_defaults_tools.xml
 - Yet your desired configuration may not be in this file
 - Make sure to name the output files appropriately
- Run mksurfddata to create the files
 - surfddata is a single-year file that contains data that is constant over time
 - the year is based on a mksurdata argument and the mksrf_fvegtyp namelist item
 - landuse.timeseries covers the years listed in the text file pointed to by the mksrf_fdynuse namelist item
- Note: to add an irrigated crop in the last PFT index the crop value needs to be split
 - This is because the LUT does not include an irrigated crop
- Copy the surfddata and landuse.timeseries files to ../inputdata/lnd/clm2/surfddata_map

~~Crop model – PFTs and CFTs~~

- set mksurfddata numpft namelist item to > 16 (bare PFT is not included in this number)
- CFT data sourced from NCAR surfddata/landuse.timeseries files
- surfddata file
 - See [Creating surface data file with increased PFTs \(50\)](#)
 - Two input files specific to these PFTs have been generated for use by mksurfddata:
 - mksrf_fvegtyp =
'/qfs/people/sinh210/wrk/user_inputdata/lnd/clm2/surfddata_map/mksrf_newcrop_c200813.nc'
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- landuse.timeseries file
 - See [Creating landuse timeseries for 50 PFTs](#)
 - PFT/CFT LUT-style files have not been created for use by mksurfddata

- Instead, previously made landuse.timeseries files have been modified to include the CFT data

~~With topounits~~

- Current workflow starts with land surface files from mksurfdata
 - python scripts to calculate topounit and associated surface variable data
 - topounit and associated surface variable data added to these files
 - two surface file types: simple and complete
- Simple surface file with topounit dimension but same surface variable data across topounits
- Complete surface file with topounit-specific data for surface variables
 - LAI, SAI, canopy height, soil properties (clay, sand, organic matter), land units, pfts/cfts, others?
- Creating topounits and associated surface variable data
 - 90m global digital elevation model
 - Determine topounits at desired grid resolution (see [topounit workflow](#))
 - Determine topounit-specific surface characteristics
 - associate 1km data (Li et al, ESSD, 2024) with topounits, spatially at 90m, and calculate averages

Current mksurfdata data files (mksurfdata namelist), v3.0.2

- area weighted averages are used to convert across grids, unless specified
 - this means that area data are calculated exactly because these values are percent/fraction of grid cell
- Note that soil data are not specific to natural veg vs crop
- The land unit/cover and land use data are desired at 1km

Data	Mapping	Description	Res	Update	Needs
mksrf_fgrid	mksrf_fgri d	Defines the output resolution grid Can be a mapping file (uses destination grid) or a grid file	0.5 degrees modis		

			to output res		
mksrf_vegtyp	map_fpft map_fharvest	PFTs/harvest - one year for fsurdat: PFT percent of veg land unit and Ffraction of land unit harvested	0.5 deg	Yes	irrigated crop and/or all specific crop types
mksrf_dynuse	map_fpft map_fharvest	PFTs/harvest time series: text file for fdyndat- list of single year files containing: PFT percent of veg land unit and Fraction of land unit harvested	0.5 deg modis	Yes	irrigated crop and/or all specific crop types
mksrf_furban	map_furban	Urban: percent of grid cell for each density class; region ID; global area of each region ID Urban params: 14 building/road scalars; 4 building/road vars by solar type (DIR/DIF) and solar band (VIS/NIR); 6 building/road vars by vertical level	3 arcmin landscan		Dynanic urban
mksrf_flakwat	map_flakwat	Lake: percent of grid cell; lake depth output lake depth is an area weighted average	3 arcmin modis		Dynamic lake
mksrf_fwetland	map_fwetland	Wetland: percent of grid cell	0.5 deg avhrr		Dynamic wetland
mksrf_glacier	map_fglacier	glacier/ice cap and ice sheet: percent of grid cell, by elevation bin	3 arcmin globe- gardner		Dynamic glacier

mksrf_fpeat	map_fpeat	Fraction of grid cell How is this used in ELM?	0.5 deg avhrr		These data should be associated with particular land units and/or PFT/CFTs
mksrf_ftopostats	map_ftopostats	grid cell elevation and slope For elev stddev and average slope of out cell	1k? or 10 arcmin?, or some of each?, 10min - hydro1 k- merge		Topography data should be consistent, suggesting only one file instead of 4. Should these topography data correspond with CAM topography? if this has variable resolution, it should probably be preprocessed to a constant res
mksrf_furbtopo	map_furbtopo	Elevation of grid cells. average value is output Used to remove urban area above elev_thresh	10 arcmin		one topo file

mksrf_flndt opo	map_flndt opo	Elevation of grid cell - for average elevation of out cell Note that this is a different data file than used for topostats	10 arcmi n		one topo file
mksrf_ftopr ad	map_ftopr ad	5 topography variables SINSL-SINAS, SINSL-COSAS, sky view, terrain config factor are unitless stddev elev (m) for top radiation model (use_top_solar_rad = .true.)	0.01 deg		one topo file; e.g., there are two stddev elev inputs
mksrf_flai	map_flai	LAI, SAI, canopy height top, canopy height bottom; monthly one-sided lai/sai, meters for canopy heights used only for satellite phenology mode is this only for FATES? (use_fates_sp = .true.)	0.5 deg avhrr		1km
mksrf_fsoic ol	map_fsoic ol	soil color class (either 20 or 8 classes, not including no soil) single layer dominant class in cell is selected, default otherwise	0.5 deg modis		
mksrf_fsoit ex	map_fsoit ex	percent sand, percent clay, igbp soil class single layer values selected by dominant valid igbp soil class in cell, up to second dominant, then set to default	5 arcmi n igbp- gsdp		1km
mksrf_fsoio rd	map_fsoio rd	soil order class (16, not including no soil) single layer	0.5 deg modis		

		dominant class in cell is selected, otherwise default (all glacier is also default)			
mksrf_forg anic	map_forg anic	organic matter density (kg/m ³) per layer, 10 layers soil layer depth and thickness (m) not used assumed carbon content 0.58 gC per gOM	5 arcmin isric-wise		1km
mksrf_fmax	map_fmax	maximum saturated grid cell fraction nodata set to global average	3 arcmin usgs		
mksrf_fgrvl	map_fgrvl	percent gravel per layer, 10 layers soil layer depth and thickness (m) not used	5 arcmin isric-wise		
mksrf_slp1 0	map_fslp1 0	slope min and decatiles (km/km): 11 values: first index is min, then 10 to 100 percentiles, in multiples of 10	0.5 degrees avhrr		
mksrf_fero	map_fero	5 erosion parameters c1 (rain hillslope), c2 (runoff hillslope), c3 (hillslope transport) are unitless these are for use_erosion = .true. : lithology erodibility is unitless fraction of grid cell under conservation tillage	0.5 deg avhrr		
mksrf_fvic	map_fvic	4 VIC parameters: b - vic curve unitless; Ws, Ds - ARNO curve unitless; Dsmax - ARNO curve mm/day	0.9x1.25 deg grdc		

mksrf_fvoc ef	map_fvoc ef	6 VOC emission factors (micrograms/m ² /h) broadleaf tree, crop, fineleaf deciduous, fineleaf evergreen, grass, shrub	0.5 deg avhrr		
mksrf_fch4	map_fch4	3 CH4 factors F0: maximum inundated gridcell fraction P3: qflx_surf_lag coefficient for finundated (s/mm) ZWT0: decay factor for finundated (m)	360x7 20 crunc ep - comp are to 0.5		
mksrf_fgdp	map_fgdp	GDP in thousand 1995US\$ per capita year 2000 CMIP5 data what is this used for?	0.5 deg avhrr	possi ble	timeseries GDP? option to use gdp in E3SM- GCAM for downscaling emissions GCAM 8.2 GDP data:  GCAM v8.2 D ocumentatio n: External In puts for Mode ling the Econ omy
mksrf_fab m	map_fabm	month of maximum agricultural fire burning based on 1997-2009 data dominant value in cell is selected	0.5 deg avhrr		
mksrf_ffert	map_ffert	Nitrogen (g/m ²) and Phosphorous (g/m ²) fertilizer per cft (36)	5 arcmi n		

		data year? for crop model (use_crop = .true. and make_crop_landunit = .true.)	isric- wise		
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Desired new mksurfddata data files

- Fine-res replacements for current data listed above
- New data for potential new features

Data	Mapping	Description	Res	Update	Needs